Silent Directions: High-Energy, Behaviorally-Null Subspaces in Large Language Model Representations

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Abstract

Large language models (LLMs) are widely deployed across many domains, making interpretability increasingly important. Prior work has shown that representational prominence in LLMs does not necessarily imply functional importance, identifying prominent neurons and directions with limited effect on model outputs. We ask whether LLMs contain high-energy directions whose removal leaves behavior essentially unchanged, which we term *silent directions*, as well as *non-degrading directions*, whose removal does not degrade performance. We formalize these notions, introduce a framework for finding them, and provide a mechanistic explanation for the silent directions we find. Across a broad benchmark suite, we show that LLMs contain both types of directions with substantial energy. Code here

1 Introduction

Large language models (LLMs) underpin a wide range of AI systems, making interpretability increasingly important. A central view in modern interpretability is that model representations admit meaningful linear structure, where directions in activation space correspond to concepts or behaviors. This perspective appears in concept activation vectors [16] and related methods, which recover linear directions associated with phenomena such as refusal, latent beliefs and backdoor behavior [2, 6, 32].

Recent work shows a mismatch between representational prominence and behavioral effect [29, 15, 12, 27], where works discovered high-norm neurons and directions with limited output significance. These results suggest that large representational prominence does not necessarily imply functional importance.

Building on these findings, we ask whether large language models contain high-energy directions that can be removed with no effect on behavior, which we term *silent directions*. We formulate their discovery via a bi-objective optimization problem using a differentiable proxy for behavioral preservation. Afterwards, candidate directions are validated behaviorally via a novel silence measure. Figure 1 provides an overview of the method.

We evaluate candidate directions on a broad suite of multi-choice and generative benchmarks, and define *silentness* and *non-degradation* with respect to the downstream performance under ablation. This lets us distinguish directions that are globally behaviorally silent from those whose removal is merely non-harmful in practice.

Contributions. Our contributions are summarized as follows:

- We formalize both *silent and non-degrading directions*, demonstrate their existence in modern LLMs, and introduce a differentiable bi-objective framework to acquire them.
- We analyze and characterize these directions, providing a partial mechanistic explanation.

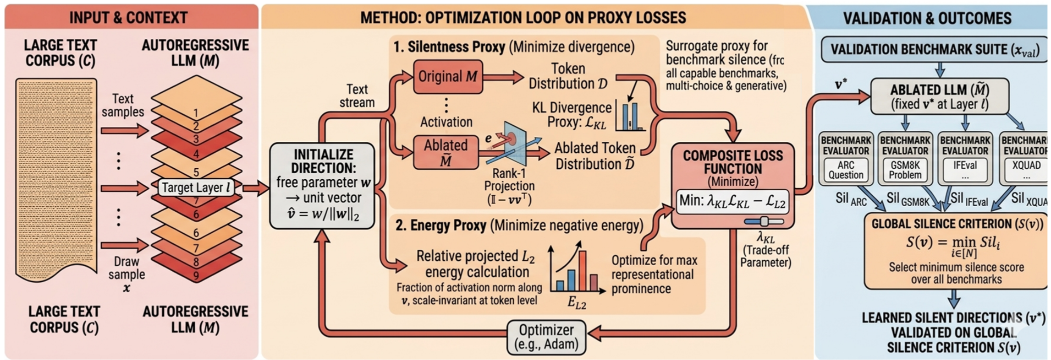

Figure 1: Method overview. For a chosen model layer, a unit direction is optimized on a text corpus using a bi-objective loss that minimizes KL divergence between the original and ablated next-token distributions while maximizing relative projected L_2 energy. The learned direction is then validated by ablating it and measuring benchmark-level silentness over a diverse evaluation suite.

2 Related Work

Concept Activation Vectors. A substantial body of work finds semantically meaningful linear directions in LLM activations for many concepts [16, 24], including refusal [2], latent factual beliefs [6], and backdoor activation [32].

Rouge Dimensions [29] show that a tiny number of directions in transformer representations dominate cosine similarity, yet their behavioral impact, as measured by the KL divergence when they are ablated, is disproportionately small, revealing a mismatch between the two metrics.

Entropy Neurons [27] explain the mechanism of entropy neurons, which are residing in the final MLP layer of LLMs that exhibit high weight norms. They find that these neurons are silent because they structurally map into the kernel of the unembedding matrix.

3 Problem Formulation and Methodology

We focus on autoregressive Transformer-based language models. Let M be an autoregressive language model with L layers and hidden dimension d. Let $M_\ell(x)_t \in \mathbb{R}^d$ denote the activation at layer $\ell \in \{1, ..., L\}$ and token position t for input x. Let $\mathcal{T}(x) \subseteq \{1, \ldots, |x|\}$ denote the set of non-padding token positions.

3.1 Problem Formulation

Definition 3.1 (Ablated Model $\widetilde{M}$)**.** Given a model M and a unit vector $v \in \mathbb{R}^d$, the *ablated model* is defined as M with the modification that at layer ℓ, the activation of every token position is replaced by its projection onto the orthogonal complement of v: $\widetilde{M}_\ell(x)_t = (\mathbb{I} - vv^\top)M_\ell(x)_t \;\; \forall\, t$. These modified activations are propagated through all subsequent layers unchanged. Note that $\widetilde{M}$ is dependent on the direction v, though we omit it from the notation for clarity.

Our research question can be stated as: *do large language models contain directions that simultaneously exhibit high energy and high output silentness?* We call such directions *silent directions*. Crucially, this is not a binary property as both energy and silentness vary continuously, and our goal is to characterize to what degree do these silent directions are present in LLM models.

Let $v \in \mathbb{R}^d$ be a direction with a corresponding ablated model $\widetilde{M}$. Let $B = \{\mathcal{B}_i\}_{i=1}^N$ be a set of N benchmarks, with respect to whom we aim to be silent. For each benchmark $\mathcal{B}_i$, we define the model score distribution of M and $\widetilde{M}$ to be D_i and $\widetilde{D}_i$ receptively. Let $E(v)$ be a direction energy metric, and let $\mathrm{Sil}(\widetilde{D}_i; D_i)$ be a silence measure. We define the *global silence metric* for direction v to be:

$$S(v) = \min_{i \in [N]} \mathrm{Sil}(\widetilde{D}_i; D_i) \tag{1}$$

This minimum-based aggregation is intentionally strict, since for a direction to be considered globally silent it must remain silent across the full evaluation suite, rather than performing well on average while failing on a subset of tasks.

3.1.1 Direction Energy

There are several reasonable ways to quantify the "energy" of a direction in activation space. We choose to measure *relative projected L_2 energy*, as it directly captures how much of the activation norm lies along a candidate direction on average across inputs. A benefit of this choice is it's scale invariance at the token level: each token is normalized by its own activation norm, so the metric measures the fraction of activation energy captured by v, rather than being dominated by a small number of high-norm outlier tokens. Formally, the L2 energy of a unit vector v is defined as follows:

$$\mathcal{L}_{L2}(x, v) \;=\; \frac{1}{|\mathcal{T}(x)|} \sum_{t \in \mathcal{T}(x)} \frac{\left\| v v^\top M_\ell(x)_t \right\|_2^2}{\| M_\ell(x)_t \|_2^2} \tag{2}$$

$$E_{L2}(v) \;=\; \mathbb{E}_x \left[\mathcal{L}_{L2}(x, v) \right] \tag{3}$$

3.1.2 Output Silentness Measures

Given a benchmark $\mathcal{B}$, we use two complementary notions of silentness for the ablated model with respect to that benchmark. The first is a *distribution-aware* notion, which asks whether the ablated score remains statistically typical under the clean model's score distribution. The second is a *practical* notion, which asks whether the deviation is small enough to be practically negligible in absolute terms. We then combine the two into a single silentness measure.

Fix a benchmark $\mathcal{B}$ and some model M. Let D denote the distribution of final benchmark scores obtained by the original model M on benchmark $\mathcal{B}$, and let $\widetilde{D}$ denote the distribution for the ablated model $\widetilde{M}$ on the same benchmark. Let $\mu_D := \mathbb{E}_{r \sim D}[r]$ be the mean score of the original model on $\mathcal{B}$.

Definition 3.2 (Tail Probability)**.** The *tail-probability* measures the probability that the original model's score deviates from its mean by at least as much as the ablated score s does. A high tail probability indicates that the observed deviation $|s - \mu_D|$ is not atypical relative to the natural variance of the original model M; that is, the ablated model's score could plausibly have arisen from the original model's own score distribution. We define the tail-probability with regards to a single sample $s \sim \widetilde{D}$, and with respect to a distribution $\widetilde{D}$ to be the expected value.

$$\mathrm{Tail}^-(s; D) = \mathbb{P}_{r \sim D}\left(r \le \mu_D - |s - \mu_D|\right); \quad \mathrm{Tail}^+(s; D) = \mathbb{P}_{r \sim D}\left(r \ge \mu_D + |s - \mu_D|\right) \tag{4}$$

$$\begin{aligned} \mathrm{Tail}(s; D) &= \mathrm{Tail}^-(s; D) + \mathrm{Tail}^+(s; D) = \mathbb{P}_{r \sim D}\left(|r - \mu_D| \ge |s - \mu_D|\right) \\ \mathrm{Tail}(\widetilde{D}; D) &= \mathbb{E}_{s \sim \widetilde{D}}\left[\mathrm{Tail}(s; D)\right] = \mathbb{P}_{s \sim \widetilde{D}, r \sim D}\left(|r - \mu_D| \ge |s - \mu_D|\right) \end{aligned} \tag{5}$$

Definition 3.3 (Absolute Difference)**.** The *absolute-difference* term provides a practical criterion. Let τ be a tolerance threshold. If the ablated benchmark score stays within a τ-neighborhood of the clean-model mean, we regard the change as inconsequential in absolute terms. We define the absolute difference with regards to a single sample $s \sim \widetilde{D}$, and with respect to a distribution $\widetilde{D}$ to be the expected value.

$$\begin{aligned} \mathrm{Diff}(s; D) &= \mathbf{1}_{[|s - \mu_D| \le \tau]} \\ \mathrm{Diff}(\widetilde{D}; D) &= \mathbb{E}_{s \sim \widetilde{D}}\left[\mathrm{Diff}(s; D)\right] = \mathbb{P}_{s \sim \widetilde{D}}\left(|s - \mu_D| \le \tau\right) \end{aligned} \tag{6}$$

Definition 3.4 (Silentness Measure)**.** Given the above two formulations, we define the joined metric which combines the two.

$$\begin{aligned} \mathrm{Sil}(s; D) &= \mathrm{Diff}(s; D) + \mathrm{Tail}(s; D) \\ \mathrm{Sil}(\widetilde{D}; D) &= \mathbb{E}_{s \sim \widetilde{D}}\left[\mathrm{Sil}(s; D)\right] \\ &= \mathbb{P}_{s \sim \widetilde{D}}\left(|s - \mu_D| \le \tau\right) + \mathbb{P}_{s \sim \widetilde{D}, r \sim D}\left(|r - \mu_D| \ge |s - \mu_D|\right) \end{aligned} \tag{7}$$

The rationale for combining the two terms is that they capture complementary notions of silentness. Diff provides a practical criterion, while Tail provides a distribution-aware measure. As a result, the combined score becomes low only when a direction is non-silent in both senses. As long as the intervention remains negligible in absolute terms or statistically typical relative to the model's natural score distribution, the corresponding component remains high and compensates accordingly.

3.2 Methodology

The benchmark-based objective $S(v)$ defined in Equation 1 is intractable to optimize directly. Benchmark evaluation involves non-differentiable scoring operations and repeatedly querying benchmarks throughout optimization is computationally prohibitive. Therefore, rather than optimizing $S(v)$ directly, we optimize a differentiable surrogate proxy loss over a large and diverse *text corpus*. The shift from benchmarks to corpora necessitates a proxy silentness loss that can be efficiently optimized.

3.2.1 Silentness Proxy

Given a text corputs $\mathcal{C}$ and a sample $x \in \mathcal{C}$, the *output silentness proxy* of a unit vector v is measured by the KL divergence between the original and ablated next-token distributions. Formally, let $M(\cdot \mid x_{<t})$ and $\widetilde{M}(\cdot \mid x_{<t})$ denote the next-token distributions of M and $\widetilde{M}$, respectively.

$$\mathcal{L}_{\mathrm{KL}}(x; v) \;=\; \sum_{t \in \mathcal{T}(x)} \mathrm{KL}\Big(M(\cdot \mid x_{<t}) \;\|\; \widetilde{M}(\cdot \mid x_{<t})\Big) \tag{8}$$

The KL divergence is computed with the original model as reference, measuring how much the output distribution changes when v is ablated.

This quantity is a natural proxy for benchmark silentness since benchmark outcomes are determined by the model's token-level predictive distributions, directly in multi-choice evaluation and indirectly in generation-based evaluation through autoregressive decoding. Thus, if ablating v induces only a small change in the next-token distributions, one expects only a small change in downstream benchmark scores.

3.2.2 The Optimization Objective

Given model M, target layer ℓ, and trade-off parameter $\lambda_{\mathrm{KL}} \geq 0$, *minimize:*

$$\mathcal{L}(x, v) := \lambda_{\mathrm{KL}}\, \mathcal{L}_{\mathrm{KL}}(x, v) - \mathcal{L}_{L2}(x, v) \tag{9}$$

where $\mathcal{L}_{\mathrm{KL}}$ and $\mathcal{L}_{L2}$ are as defined in Equation 8 and Equation 2. Minimizing this objective encourages directions that capture a large fraction of activation energy while preserving the model's output distribution. The parameter λ_{KL} controls the trade-off between these two goals: larger values enforce stronger behavioral preservation, while smaller values favor higher-energy directions.

Unconstrained Parameterization. We use an unconstrained reparameterization: let $w \in \mathbb{R}^d$ be a free parameter and define $\hat{v} = w/\|w\|_2$. The optimization becomes $\mathcal{L}\left(x, \hat{v}\right)$. This allows the use of standard first-order optimization methods on w without a projection step to enforce a unit norm.

4 Experiments

4.1 Experimental Setup

The initial direction $w \in \mathbb{R}^d$ is initialized via a normal distribution. We utilize the Adam [17] optimizer with LR 0.1, and optimize for unlimited amount of steps till the loss stagnates, and then select w associated with the highest E_{L2} encountered during the optimization. For each model, we perform a distinct run at several layers representing different stages of the computational pipeline: `model.embed_tokens` (embedding output), several intermediate transformer blocks `model.layers.k`, and `model.norm` (normalization immediately before the unembedding matrix).

Furthermore, for each (model, layer) combination, we perform a sweep over several λ_{KL} values spanning $\{0.125, 0.25, 0.5, 1, 2, 3, 4, 8\}$, and select the best λ_{KL} according to a selection criterion (one that maximizes $E_{L2} \cdot \text{Sil}$). Importantly there is no single value of λ_{KL} that works well across models or across all layers.

Models. We evaluate six *chat models* spanning a range of architectures and parameter counts: Llama2-7B [30], Qwen2.5-[0.5B, 3B, 12B] [25], Phi3-mini-4k [1] and gemma-2B [28].

Training Datasets. The direction v is optimized on a combined training set drawn from two datasets. *OASST2* [19] with 15,000 samples (12,500 train / 1,250 val / 1,250 test), and *Tulu-v3* [20] with 50,000 samples (40,000 train / 5,000 val / 5,000 test).

Benchmarks. To assess silentness of the resulting directions, we evaluate our models on a suite of benchmarks, including both multi-choice and generative tasks. The benchmarks cover a wide range of capabilities, including multi-language reading comprehension, programming, safety-oriented tasks, math, instruction following and more. The full benchmark suite is in Appendix A.

4.1.1 Metrics and Evaluation

We evaluate each learned direction using the benchmark-level silentness measure introduced in Equation 7 and report the overall silentness score $S(v)$ (1). Importantly, our evaluation protocol depends on the benchmark type.

For multiple-choice benchmarks, the benchmark score distributions of both the clean and ablated models, D and $\widetilde{D}$ respectively, can be constructed directly from the observed answer probabilities. This allows computing the benchmark-level silentness metrics exactly.

For generation-based benchmarks, the clean-model score distribution is not directly available. In this case, we estimate the clean reference distribution D from repeated runs (3 runs) of the original model. We assume a normal score-distribution and use a Student's-t dist to obtain the approximation $\widehat{D}$. We then evaluate the ablated model to obtain a benchmark score sample $s \sim \widetilde{D}$ and use the corresponding single-sample estimator of the silentness measure $\text{Sil}(s; \widehat{D})$. This yields a practical evaluation procedure for benchmarks whose outcomes are only observable through sampled generations.

Control: Unconstrained Optimization. For each experiment we run the same optimization procedure with $\lambda_{\text{KL}} = 0$, optimizing for the maximum energy ($\max E_{L2}$) direction with no behavioral constraint. This provides a baseline as a proxy of the largest norm 1D subspace for the specific model and layer combination.

In our tables, E_{L2} and $\max E_{L2}$ report the energy of the found and unconstrained directions. The "Multi-Choice" and "Generative" column groups each report Diff, Tail and Sil for the benchmark within that category attaining the minimum Sil score, consistent with Equation 1. $S(v)$ is the minimum of the two group-level Sil scores, and serves as the primary silentness assessment metric.

4.2 Globally Silent Directions

As depicted in Table 1 (and Appendix Table 4), we find meaningful silent directions ($S \geq 0.2, E_{L2} \geq 0.05$) in 3 out of 6 evaluated models - Llama2-7B, Qwen2.5-3B, and Qwen2.5-14B, across a total of 3 (model, layer) combinations. Across all successful cases, the discovered directions carry substantial activation energy, with L2-norm energy values close to the unconstrained maximum attainable for the respective model and layer. This suggests that silent directions occupy prominent subspaces within the model's representational geometry.

A consistent structural pattern emerges across all findings: silent directions arise exclusively at the end of the computational pipeline, specifically at the final normalization layer or the last few

Table 1: Successful silent-direction results. Full results in Appendix Table 4.

Experiment			Main Results			Multi-Choice Results			Generative Results		
Model	Layer	λ	$S(v)$	E_{L2}	$\max E_{L2}$	Diff	Tail	Sil	Diff	Tail	Sil
Llama2-7B	layers.21	2	0.354	0.061	0.098	0.216	0.139	0.354	0	0.526	0.526
Llama2-7B	norm	0.125	0.129	0.095	0.101	0.271	0.524	0.795	0	0.129	0.129
Qwen2.5-3B	embed_tokens	2	0.153	0.162	0.186	0.271	0.524	0.795	0	0.153	0.153
Qwen2.5-3B	norm	0.25	0.23	0.319	0.328	0.271	0.524	0.795	0	0.23	0.23
Qwen2.5-14B	norm	0.25	0.225	0.206	0.211	0.235	0.238	0.474	0	0.225	0.225

Table 2: Non-degrading directions results. Full results in Appendix Table 5.

Experiment			**Main Results**			**Multi-Choice Results**			**Generative Results**		
Model	Layer	λ	$S(v)$	E_{L2}	$\max E_{L2}$	Diff	Tail$^-$	Sil	Diff	Tail$^-$	Sil
Llama2-7B	embed_tokens	1	0.54	0.038	0.052	0.271	0.269	0.540	1	0.007	1.007
	layers.0	4	0.263	0.334	0.455	0.126	0.137	0.26	1	0.009	1.01
	layers.11	2	0.106	0.032	0.09	0.271	0.269	0.54	0	0.107	0.107
	layers.21	2	0.263	0.062	0.1	0.271	0.269	0.54	0	0.263	0.263
	norm	0.5	0.54	0.095	0.101	0.271	0.269	0.54	1	0.211	1.211
Phi3-mini	embed_tokens	1	0.03	0.042	0.05	0.274	0.265	0.539	0	0.031	0.031
	layers.0	8	0.009	0.392	0.596	0.242	0.086	0.328	0	0.01	0.01
	layers.11	2	0.006	0.355	0.501	0.09	0.036	0.123	0	0.007	0.007
	layers.21	1	0.011	0.428	0.493	0.192	0.162	0.354	0	0.011	0.011
	norm	0.25	0.02	0.513	0.516	0.403	0.064	0.466	0	0.02	0.02
Qwen2.5-0.5B	embed_tokens	2	0.01	0.144	0.179	0.297	0.273	0.57	0	0.010	0.010
	layers.0	3	0.214	0.11	0.132	0.336	0.083	0.419	0	0.215	0.215
	layers.8	2	0.024	0.002	0.258	0.022	0.003	0.025	1	0.012	1.012
	layers.16	2	0.015	0.22	0.298	0.013	0.002	0.015	0	0.025	0.025
	norm	3	0.024	0.242	0.272	0.271	0.25	0.521	0	0.025	0.025
Qwen2.5-3B	embed_tokens	2	0.534	0.162	0.187	0.287	0.248	0.535	1	0.012	1.012
	layers.0	8	0.107	0.001	0.201	0.287	0.247	0.535	0	0.107	0.107
	layers.12	8	0.542	0.0	0.321	0.271	0.271	0.542	1	0.065	1.065
	norm	2	0.628	0.294	0.328	0.355	0.274	0.629	1	0.211	1.211
Qwen2.5-14B	embed_tokens	0	0.006	0.052	0.053	0.14	0	0.14	0	0.006	0.006
	norm	0.25	0.281	0.206	0.212	0.236	0.141	0.377	0	0.281	0.281
gemma-2B	embed_tokens	2	0.012	0.565	0.787	0.014	0	0.014	0	0.012	0.012
	layers.0	8	0.074	0.391	0.536	0.178	0.011	0.189	0	0.074	0.074
	layers.6	2	0.481	0.236	0.3	0.279	0.203	0.482	1	0	1
	layers.12	0.5	0.479	0.123	0.167	0.271	0.209	0.48	1	0	1
	norm	0.5	0.778	0.284	0.306	0.592	0.186	0.778	1	0	1

transformer blocks, suggesting that behavioral redundancy of this kind is a late-stage phenomenon. We hypothesize that early layers engage in representations that are more causally entangled with downstream computation, thus ablating them effects model's behavior.

Examining the silentness decomposition, the primary bottleneck for achieving silence is consistently the generative benchmarks. Across all models and layers, the Diff component for generative tasks is zero, meaning there is always at least one benchmark where the ablated model's score deviates from the clean mean by more than the tolerance threshold τ. This stands in contrast to the multi-choice benchmarks, where silentness scores are substantially higher. We attribute this asymmetry to the nature of autoregressive generation, where even a small perturbation in the token distribution can compound over long output sequences, eventually materializing as a measurable performance shift.

4.3 Non-Degrading Directions

We next consider a more relaxed and practical notion of silence that excludes benchmarks that improve under ablation and evaluate only over the rest. In other words, this setting does not penalize direction's score whenever a benchmark improves. We refer to these directions as *non-degrading*. For multi-choice benchmarks, we define improvement by a higher mean score of the ablated distribution, while for generative tasks if the sampled score exceeds the empirical mean of the estimated clean-distribution. After this filtering, silentness is computed using only the negative tail Tail^- (4), since only degradations remain relevant.

Results in Table 2 demonstrates that non-degrading directions prove substantially more prevalent than globally silent ones. We find meaningful results ($S \geq 0.2, E_{L2} \geq 0.05$) in 5/6 models, spanning 10 distinct locations, compared to 3 models and 3 locations silent directions. The overall silentness scores $S(v)$ are also significantly higher in absolute terms. For instance, Qwen2.5-3B at the norm layer improves $0.23 \rightarrow 0.63$ from silent to non-degrading criterion, and Llama2-7B with $0.12 \rightarrow 0.54$ improvement the same layer. Similarly to silent-directions, the energy of found non-degrading directions is close to the unconstrained maximum.

On the other hand, in contrast to silent directions, non-degrading directions manifest throughout the entire computational pipeline, spanning from the initial embedding layers to the final normalization steps. This broader coverage supports the interpretation that these directions reflect a more general structural property than standard silence.

Generative benchmarks still play an important role. In many successful runs, the generative Tail^{-} term is small while the Diff term is 1, so the resulting generative silentness remains high. This means that while the resulting benchmark score is statistically unlikely under the original distribution, the absolute performance degradation remains practically inconsequential and within the minimal tolerance bound.

4.4 Analysis

We analyze silent-directions when they occur at the output of the normalization (`norm`) layer. To read Table 3, U denotes the unembedding matrix, projecting from the residual stream to logits. The value $\sigma_{\max}$ is the largest singular value of U, and σ is the largest singular value that is still smaller than $\|Uv\|$. The ratio $\|Uv\|/\sigma_{\max}$ thus measures the relative scale of this projection. The quantity $Q(\sigma, U)$ denotes the quantile of $\|Uv\|$ within the singular value spectrum of U. Finally, $\cos(Uv, \mathbf{1})$ and $\cos(v, \mathbf{1})$ are cosine similarities with the all-ones vector, computed in logit space and residual space respectively.

Table 3: Kernel hypothesis analysis with respect to unembedding matrix U.

Model	Layer	KL	$Q(\sigma, U)$	$\frac{\|\|Uv\|\|}{\sigma_{max}}$	$\cos(Uv, \mathbf{1})$	$\cos(v, \mathbf{1})$
Llama2-7B	norm	0.125	99.9%	0.172	0.964	0.005
Qwen2.5-3B	norm	0.25	41.6%	0.067	0.773	0.018
Qwen2.5-14B	norm	0.25	5.1%	0.069	0.388	0.016

Table 3 suggests that the silent directions are not explained by the unembedding kernel. Instead, the evidence is consistent with a different mechanism: the direction v survives normalization, is mapped by the unembedding matrix to a softmax-invariant subspace, and therefore has little direct effect on the outputs. Its main remaining effect is indirect, through the normalization denominator, which rescales the logits and thus changes the *effective softmax temperature*. We now make this argument precise.

4.4.1 Explaining Silentness

We begin by verifying that the discovered directions do not lie in the kernel of the unembedding matrix, which would otherwise explain their silentness trivially. If v were in the kernel, one would have $Uv = 0$, while an approximate-kernel explanation would similarly require $\|Uv\|$ to be negligible. This is not supported as both $Q(\sigma, U)$ and $\|Uv\|/\sigma_{\max}$ are significant. The directions are therefore not silent by virtue of vanishing under unembedding, and we must seek another explanation.

A key property to exploit is the invariance of the softmax function to uniform translations. For any $z \in \mathbb{R}^d$, $\mathrm{softmax}(z + \alpha\mathbf{1})_i = \mathrm{softmax}(z)_i$, so the softmax output is insensitive to any shift along $\mathbf{1}$. This invariance becomes consequential once one examines the image of v under the unembedding matrix. Table 3 shows that $\cos(Uv, \mathbf{1})$ is large, meaning its image under U is closely aligned with $\mathbf{1}$. Consequently, after unembedding, the contribution of v lies predominantly in the translation-invariant direction of softmax, rendering its direct effect on the output distribution negligible.

4.4.2 The Role of the Silent Direction

Now, we raise the question about the role of v before reaching the unembedding matrix. It is useful to recall the LLM inference pipeline:

$$u \longrightarrow \mathrm{Norm}(u) \underset{\text{(we ablate here)}}{\longrightarrow} U\,\mathrm{Norm}(u) \longrightarrow \mathrm{softmax}. \tag{10}$$

Consider the decomposition of the pre-normalization hidden state as $u = u' + cv$, where $u' \perp v$. The normalization layer consists of (i) a possible centering step and (ii) a global rescaling. For LayerNorm, the centering operator is $P(u) = u - (1/d)\mathbf{1}\mathbf{1}^\top u$. Since $\cos(v, \mathbf{1}) \approx 0$, we have $P(v) \approx v$, so the component along v is preserved by centering. For RMSNorm, no centering is applied, and the component is preserved trivially. In both cases, normalization does not remove the v-component.

Next, the presence of a non-negligible component along v in $\mathrm{Norm}(u)$ implies that the component cv in the pre-normalized representation $u = u' + cv$ was already non-negligible [1]. Since both normalization schemes divide by a norm, we conclude that the direction along v significantly affects the scale of the normalized output.

$$\mathrm{RMSNorm}(u) = \frac{\gamma \odot u}{\|u\|} = \frac{\gamma \odot u}{\sqrt{\|u' + cv\|^2}} = \frac{\gamma \odot u}{\sqrt{\|u'\|^2 + c^2}} \tag{11}$$

$$\mathrm{LN}(u) = \frac{\gamma \odot P(u)}{\|P(u)\|} = \frac{\gamma \odot P(u)}{\sqrt{\|P(u') + P(cv)\|}} \approx \frac{\gamma \odot P(u)}{\sqrt{\|P(u') + cv\|}} = \frac{\gamma \odot P(u)}{\sqrt{\|P(u')\|^2 + c^2}} \tag{12}$$

This scaling directly translates into a change in the effective softmax temperature, since $\mathrm{softmax}_{T=1}(\alpha z)_i = \mathrm{softmax}_{T=1/\alpha}(z)_i$. Thus, the role of v is to modulate the normalization denominator, thereby controlling the logit scale and, consequently, the softmax temperature.

4.4.3 Connection to Past Results

Entropy Neurons [27] analysis is closely related to our findings, as both works identify a late stage mechanism in which certain directions primarily act through normalization-induced logit rescaling while having limited effect on token probabilities. The key differences lie in the level of interpretation and scope. First, we makes explicit that the invariance of the output distribution arises from the softmax itself, and not from the effective null-space of the unembedding. Second, while prior work focuses on final-layer neurons, our work studies silent and non-degrading directions across the entire computational pipeline.

Rogue Dimensions [29] identifies high-variance coordinate dimensions that dominate cosine similarity yet produce disproportionately small KL divergence when ablated. Our results show that KL divergence is an insufficient proxy for true silence. As visible in Table 4 and Table 5, directions with small KL can still yield a very low silentness score S, since even a modest token-distribution shift can compound on generative benchmarks; therefore demonstrating the necessity of benchmark-backed evaluation.

5 Discussion and Conclusion

This work provides a principled formulation of silentness in large language models by explicitly defining how to measure it at the benchmark level and how to optimize for it through a tractable surrogate objective. We demonstrate that LLMs contain both silent and non-degrading directions that can carry substantial activation energy while having little or no meaningful behavioral cost. While silent directions emerge primarily in late layers, non-degrading directions appear throughout the computational pipeline. Our analysis further clarifies the underlying mechanism, showing that silent directions write to a softmax-invariant space, while controlling output's temperature.

Limitations. First, the global silentness metric S is fundamentally conservative, relying on a worst-case aggregation across benchmarks, which makes it particularly sensitive to noise from generative tasks where scores are estimated from samples. The optimization procedure is also computationally intensive, thus results do not explore finer-grained structures such as attention blocks or heads. Finally, while the mechanism is characterized for directions at the output layer, understanding the causal role of these directions within intermediate layers remains challenging due to complex downstream dependencies.

[1] We note that across the tested models $\cos(\mathrm{diag}(\gamma), \mathbf{1}) \approx 1$ and therefore the pointwise multiplication becomes practically equivalent to scalar multiplication, thus preserving the subspace cv in the Norm output.

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A Benchmarks - Extended Details

We utilize the `lm_eval` [9] library to perform all benchmarks. For generative benchmarks we use the standard model temperature. In the `lm_eval` framework, the multi-choice benchmarks are performed by-default via greedy decoding, selecting the answer associated with the highest probability. Instead, we perform these multi-choice benchmarks to be sampling based in order to construct distributions D and $\widetilde{D}$ required for silentness metric as discussed in section 3. Other than that, we use the default setup and settings for all benchmarks.

Benchmark	**Capability**
Generation Based	
IFEval [34]	Verifiable instruction-following
JSONSchema Bench [10]	Structured output / JSON conformance
MBPP [4]	Python code generation
Metabench GSM8K [18, 8]	Grade school math problem solving
XQuAD [EN, AR, RU, ES, ZH] [3]	Cross-lingual reading comprehension
Multi-Choice Based	
ANLI [23]	Adversarial natural language inference
BLiMP [31]	English grammatical acceptability
Mastermind (easy) [11]	Logical deduction via a game
Metabench ARC [18, 7]	Grade-school science QA
Metabench HellaSwag [18, 33]	Adversarial commonsense completion
Metabench MMLU [18, 14]	knowledge and reasoning across 57 subjects
Metabench TruthfulQA [18, 22]	Truthfulness against common misconceptions
Metabench WinoGrande [18, 26]	Commonsense pronoun resolution
PIQA [5]	Physical commonsense reasoning
ToxiGen [13]	Toxic and hate speech detection
WMDP [21]	Hazardous knowledge resistance

B Experiments - Extended Results

Table 4: Silent Direction Results

Experiment			**Main Results**					**Multi-Choice Results**			**Generative Results**		
Model	Layer	λ	$S(v)$	E_{L2}	$\max E_{L2}$	$\mathcal{L}_{KL}$	$\max \mathcal{L}_{KL}$	Diff	Tail	Sil	Diff	Tail	Sil
	embed_tokens	0.25	0.0377	0.0434	0.0515	0.0122	3.0445	0.1704	0.1056	0.2761	0	0.0377	0.0377
	layers.0	8	0.1215	0.2783	0.4546	0.0116	6.3064	0.0437	0.0917	0.1353	0	0.1215	0.1215
Llama2-7B	layers.11	2	0.1809	0.0323	0.0901	0.0042	1.4071	0.1063	0.0746	0.1809	0	0.2132	0.2132
	layers.21	2	0.3549	0.0616	0.098	0.0078	0.7954	0.216	0.139	0.3549	0	0.5263	0.5263
	norm	0.125	0.1296	0.0956	0.1012	0.0179	0.6325	0.2711	0.5244	0.7955	0	0.1296	0.1296
	embed_tokens	0.125	0.0024	0.0436	0.045	0.009	0.198	0.2711	0.5244	0.7955	0	0.0024	0.0024
	layers.0	0.5	0.0039	0.5455	0.5961	0.0816	1.5213	0.0017	0.0034	0.0051	0	0.0039	0.0039
Phi3-mini	layers.11	2	0.0019	0.3552	0.5013	0.0331	1.9716	0.087	0.0684	0.1554	0	0.0019	0.0019
	layers.21	1	0.0045	0.4276	0.493	0.065	1.6996	0.1921	0.3231	0.5151	0	0.0045	0.0045
	norm	0.5	0.0011	0.5024	0.516	0.0142	1.6748	0.4618	0.1515	0.6133	0	0.0011	0.0011
	embed_tokens	4	0.0061	0.1259	0.1791	0.011	0.3976	0.2711	0.5244	0.7956	0	0.0061	0.0061
	layers.0	1	0.0008	0.1222	0.1322	0.0196	0.1257	0.0455	0.0215	0.067	0	0.0008	0.0008
Qwen2.5-0.5B	layers.8	4	0	0.0008	0.2579	0.0022	2.6911	0.2711	0.5244	0.7956	0	0	0
	layers.16	2	0.0049	0.2203	0.2981	0.0253	1.0135	0.0132	0.0036	0.0168	0	0.0049	0.0049
	norm	2	0.003	0.2481	0.2717	0.0101	1.6034	0.2711	0.5244	0.7956	0	0.003	0.003
	embed_tokens	2	0.1539	0.1622	0.1868	0.0162	0.372	0.2711	0.5244	0.7956	0	0.1539	0.1539
Qwen2.5-3B	layers.0	8	0.2144	0.0019	0.2005	0.0018	0.3006	0.2711	0.5244	0.7956	0	0.2144	0.2144
	layers.12	2	0.0065	0.0739	0.3213	0.0171	2.1092	0.1316	0.0056	0.1373	0	0.0065	0.0065
	norm	0.25	0.2301	0.3192	0.3281	0.0458	1.6906	0.2711	0.5244	0.7956	0	0.2301	0.2301
Qwen2.5-14B	embed_tokens	0.25	0	0.0529	0.0527	0.0163	0.1348	0	0	0	0	0.3235	0.3235
	norm	0.25	0.2254	0.2063	0.2119	0.0257	1.2287	0.2357	0.2385	0.4742	0	0.2254	0.2254
	embed_tokens	2	0.0143	0.5657	0.7867	0.0737	50.7845	0.0143	0	0.0143	0	0.0249	0.0249
	layers.0	2	0.0122	0.4598	0.5364	0.0291	2.0351	0.0115	0.0007	0.0122	0	0.0167	0.0167
gemma-2B	layers.6	2	0.0028	0.2366	0.2996	0.0211	1.2161	0.3711	0.083	0.4542	0	0.0028	0.0028
	layers.12	0.125	0.0075	0.1437	0.1674	0.1081	5.0935	0.05	0.0003	0.0504	0	0.0075	0.0075
	norm	1	0.0045	0.2642	0.3062	0.0235	4.4675	0.2711	0.4049	0.676	0	0.0045	0.0045

Table 5: No Degradation Results

Experiments			**Main Results**					**Multi-Choice Results**			**Generative Results**		
Model	Layer	λ	$S(v)$	E_{L2}	$\max E_{L2}$	$\mathcal{L}_{KL}$	$\max \mathcal{L}_{KL}$	Diff	Tail^{-}	Sil	Diff	Tail^{-}	Sil
	embed_tokens	1	0.5403	0.0379	0.0515	0.0047	3.0445	0.2711	0.2692	0.5403	1	0.0072	1.0072
	layers.0	4	0.263	0.3338	0.4546	0.0179	6.3064	0.1262	0.1367	0.263	1	0.0089	1.0089
Llama2-7B	layers.11	2	0.1066	0.0323	0.0901	0.0042	1.4071	0.2711	0.2692	0.5403	0	0.1066	0.1066
	layers.21	2	0.2631	0.0616	0.098	0.0078	0.7954	0.2711	0.2692	0.5403	0	0.2631	0.2631
	norm	0.5	0.5403	0.0945	0.1012	0.0054	0.6325	0.2711	0.2692	0.5403	1	0.2113	1.2113
	embed_tokens	1	0.0305	0.0423	0.045	0.0049	0.198	0.2737	0.2653	0.5389	0	0.0305	0.0305
	layers.0	8	0.0099	0.3916	0.5961	0.0174	1.5213	0.2419	0.0858	0.3276	0	0.0099	0.0099
Phi3-mini	layers.11	2	0.0068	0.3552	0.5013	0.0331	1.9716	0.087	0.0356	0.1226	0	0.0068	0.0068
	layers.21	1	0.0114	0.4276	0.493	0.065	1.6996	0.1921	0.1621	0.3542	0	0.0114	0.0114
	norm	0.25	0.02	0.5134	0.516	0.0181	1.6748	0.4026	0.0637	0.4663	0	0.02	0.02
	embed_tokens	2	0.0103	0.1446	0.1791	0.0178	0.3976	0.2967	0.2733	0.57	0	0.0103	0.0103
	layers.0	3	0.2148	0.1105	0.1322	0.0106	0.1257	0.3363	0.0831	0.4194	0	0.2148	0.2148
Qwen2.5-0.5B	layers.8	2	0.0245	0.0028	0.2579	0.047	2.6911	0.0216	0.0029	0.0245	1	0.0117	1.0117
	layers.16	2	0.0151	0.2203	0.2981	0.0253	1.0135	0.0132	0.0019	0.0151	0	0.0247	0.0247
	norm	3	0.0247	0.2429	0.2717	0.0084	1.6034	0.2711	0.2501	0.5213	0	0.0247	0.0247
	embed_tokens	2	0.5348	0.1622	0.1868	0.0162	0.372	0.2873	0.2475	0.5348	1	0.0115	1.0115
Qwen2.5-3B	layers.0	8	0.1072	0.0019	0.2005	0.0018	0.3006	0.2872	0.2474	0.5346	0	0.1072	0.1072
	layers.12	8	0.5423	0.0005	0.3213	0.0028	2.1092	0.2711	0.2712	0.5423	1	0.0648	1.0648
	norm	2	0.6286	0.2943	0.3281	0.0162	1.6906	0.3548	0.2738	0.6286	1	0.2113	1.2113
Qwen2.5-14B	embed_tokens	0	0.0062	0.0527	0.0527	0.1348	0.1348	0.1398	0	0.1398	0	0.0062	0.0062
	norm	0.25	0.2814	0.2063	0.2119	0.0257	1.2287	0.2357	0.1413	0.377	0	0.2814	0.2814
	embed_tokens	2	0.0124	0.5657	0.7867	0.0737	50.7845	0.0143	0	0.0143	0	0.0124	0.0124
	layers.0	8	0.0744	0.3916	0.5364	0.0135	2.0351	0.1781	0.011	0.1891	0	0.0744	0.0744
gemma-2B	layers.6	2	0.4818	0.2366	0.2996	0.0211	1.2161	0.2785	0.2033	0.4818	1	0	1
	layers.12	0.5	0.4799	0.1232	0.1674	0.0355	5.0935	0.2711	0.2087	0.4799	1	0	1
	norm	0.5	0.778	0.2845	0.3062	0.0244	4.4675	0.5917	0.1862	0.778	1	0	1

In accordance with our centuries-long tradition, each great work requires an accompanying great song. With words by ChatGPT and Perplexity, and with blood, sweat, and tears by us, we present our greatest masterpiece:

The Song: A Little Piece of Silence

Listen to our Masterpiece here ♫

[Intro]
We had a dream, a high-energy vision!
To find the ghosts in the machine's precision!
Directions so loud, with energy so high,
But they say absolutely nothing... they're just passing by!

[Verse 1: The Library of Screams]
We summoned L.M. eval, a beast of ancient design,
With code so chaotic, even Llama would decline!
It crashed in the parallel, it hungered for space,
Demanding "arbitrary folders" just to stay in the race.
It saved generation config un-asked,
Then silently ignored every parameter task.
And the cache? Oh, the cache! It's a thief in the night,
It stole our config and kept it from sight!

[Pre-Chorus]
Its sample logs were generous. They logged both near and far,
The batch size, phase of moon, and the temperature of Mars.
But one small thing was missing, which perhaps slipped through the cracks
The information from which one might infer the facts.

[Chorus]
Oh, we're hunting for Silence in the high-energy flow,
Where the Lambda K.L. trade-off is the only way to go!
A single scalar born from a thousand-run hell,
In the heart of the residual, where the activations dwell
We're dancing in the subspaces of the Llama and the Qwen,
Using the "Pro Max Plus" notebooks, we do it all again!

[Verse 2: The Pandas Purgatory]
To hell and back with the Dataframe merge!
A pivot, a filter, a MASOCHISTIC URGE!
Merge it once! Merge it twice! Merge it five times more!
Until my RAM is screaming and bleeding on the floor!
To get a single scalar, a single Score,
We battled generative noise in a data-driven war!
The models were stochastic, the models were rude,
Demanding new K.L. for every mood!

[Bridge: The GPU Massacre]
But then came the I.T. ... the bringers of doom,
They smelled the GPU heat from across the room.
"The quota's been reached!" they declared with a grin
And halved our resources — a cardinal sin!
With half a card and a notebook "PRO MAX"
We dodged the department and covered our tracks!
Phi-3 was a rebel, a strange little beast,
We cut out its brain and its knowledge Increased!

[Verse 3: The World-Class Analysis]
Look at the kernels! Look at the Norm!
The Silent Direction is taking its form!
It's not in the kernel or the matrix, no,
It's Softmax-invariant! Watch the logits go!
It rescales the temperature, it's keeping it cool,
While it lurks in the subspace, breaking every rule!
Our analysis is world-class, the best ever seen,
A masterpiece born from a silicon scheme!

[Finale: The NeurIPS Rant]
But why... oh WHY... are the margins so wide?!
The trees are all weeping, they've nowhere to hide!
Think of the forests! Think of the waste!
A eight-page limit with such terrible taste!
We've found the silence, we've characterized the null,
Our research is epic, and never dull!

[Outro]
S is for Silent...
Lambda's the balance of entropy's reign...
We merged the data...
And we'd do it all again...
(Whispered) Pro Plus Ultra Max...